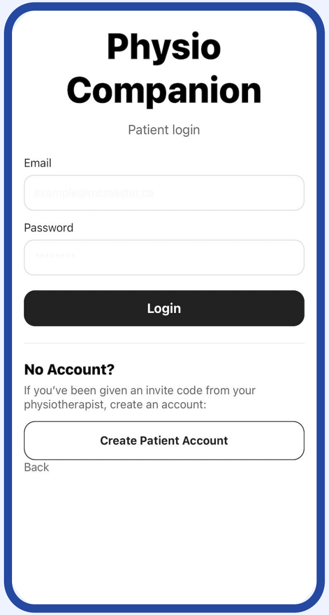
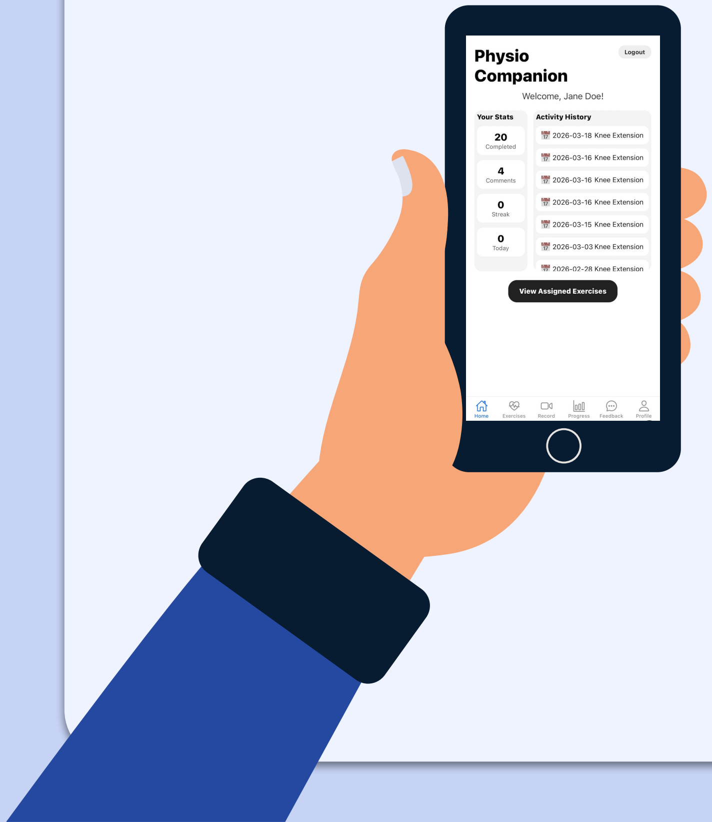
Physio Companion

DEMO



Introduction

PhysioCompanion is a mobile **physiotherapy assistant** that provides **real-time feedback** and **performance tracking** for knee rehabilitation.



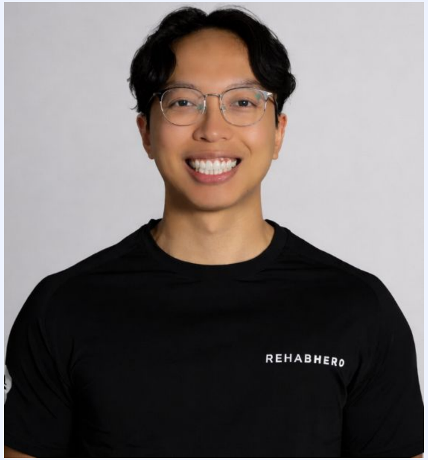
Motivation

- Lack of real-time guidance outside in-person clinic sessions
- **Goal:** To provide **real-time guidance**, helping users **stay consistent**, and ultimately supporting a more **effective recovery process** at home

Our Supervisors

- MScPT, McMaster University
- Specializes in orthopedic and knee rehabilitation
- Experience with athletes and post-operative patients

Kevin Yu (Synergy Sports Medicine, Toronto)

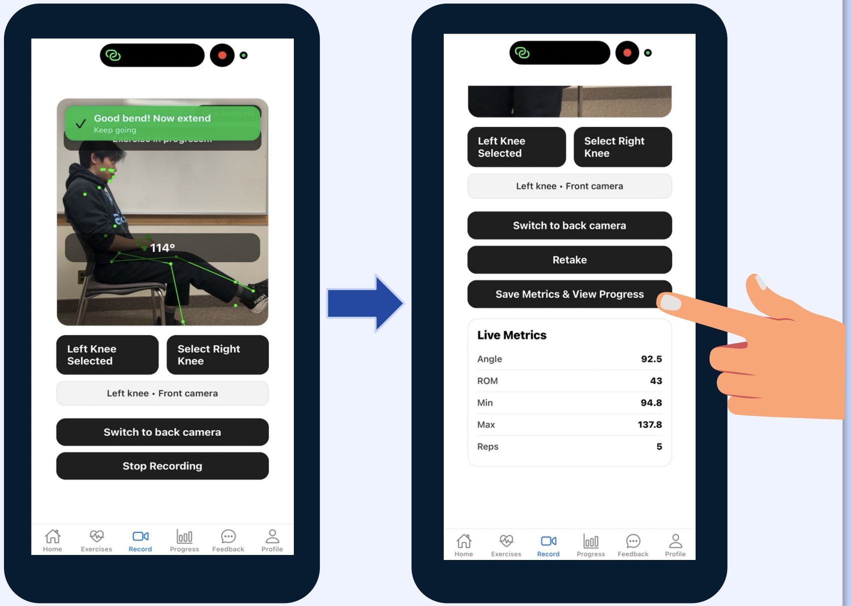


- MScPT, Queen's University
- Works with diverse patient populations
- Patient-centered approach to rehabilitation

Luke Nguyen (Rehab Hero, Don Mills)

Exercise Recording Feature

- **Frames captured** and sent to **Flask backend**
- **OpenCV** decodes and validates each frame
- **MediaPipe Pose Detection** model detects body landmarks **(hip, knee, ankle)**
- Key performance metrics **(repetition count, angle, range of motion)** are computed and displayed in **real-time**



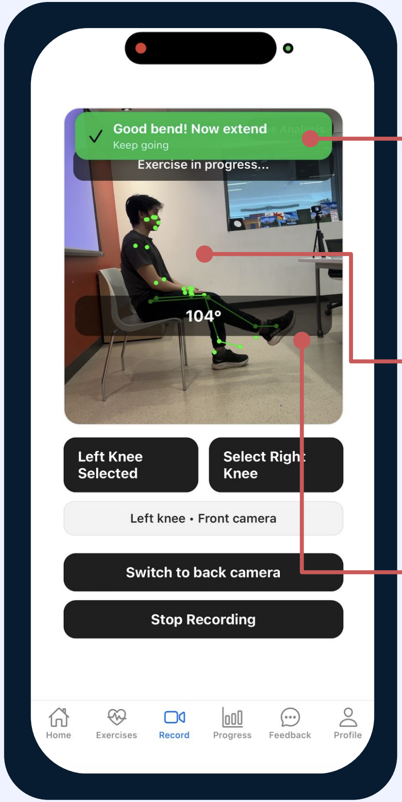
Feedback Features

Widely Accessible

Written values, visual cues, and **audio cues** provide feedback in many forms

Variable Detail

Users can dive into the **detail they desire**, reviewing metrics to simply relying on audio cues



Visual Cues

Encouraging wording and active color

Landmarks

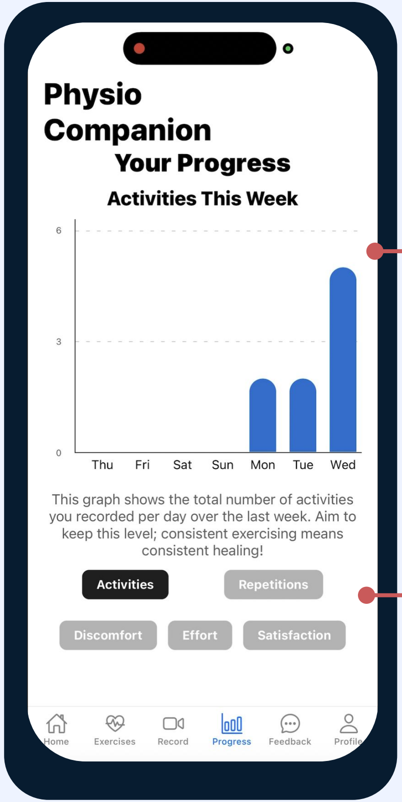
Visual representation of pose detection

Metrics

Values extracted by MediaPipe displayed for clarity

Progress Page

- Visual overview of **recovery progress**
- Tracks **consistency** in exercise completion
- Supports both **patient self-monitoring** and **physiotherapist review**



Weekly Metrics

Visual representation of weekly progress

Progress Graphs

Categorical breakdown for key progress graphs

Usability Testing

2 PTs

‘It would be highly helpful to track patient effort levels, specifically RPE (Rate of Perceived Exertion), and pain levels on a scale of 0 to 10. This is especially crucial for post-op patients to ensure they aren't pushing too hard and causing excruciating pain.’

3 patients

’One thing I noticed was that the recording feature sometimes had trouble detecting the movement, and it was a bit difficult for me to see the screen clearly when standing further away doing the exercise. ‘

Usability Testing Results

2 PTs

‘A solid 8.5.’ - when asked about ease of use.

3 patients

’Everything is laid out clearly and it’s simple to understand what to do next, which is important especially for someone recovering from a surgery. I also liked how straightforward the instructions were and that the app makes physiotherapy exercises feel more engaging instead of repetitive.’

Future Goals

- Improved Pose Detection Accuracy
  - Limitations with current MVP
- Expanded Exercise Library
  - Upper-body Exercises
  - Lower-body Exercises
- Gamification & Engagement
  - Achievement badges
  - Recurring Reminders

Conclusion

There is a need:

- Consistency
- Clarity
- Confidence

Without a solution:

- Repeat Injuries
- Repeat Expenses
- Wasted time and money

There is a way:

PhysioCompanion

- Exercise Recording
- Patient and Physio Connection

Thanks!

Questions?